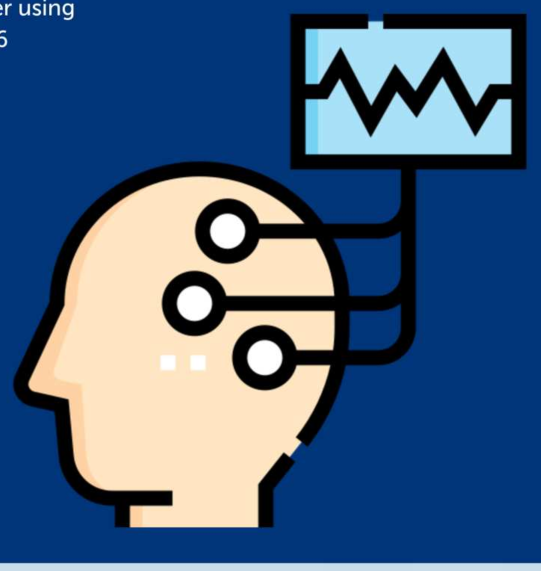
Icon by Researcher using Canva, 2026



Visual by Researcher using Canva, 2026

EpiSight

A Graph Convolutional Network for Epileptogenic Zone Localization in Drug-Resistant Epilepsy Using Phase-Amplitude Coupling

FRAMEWORK

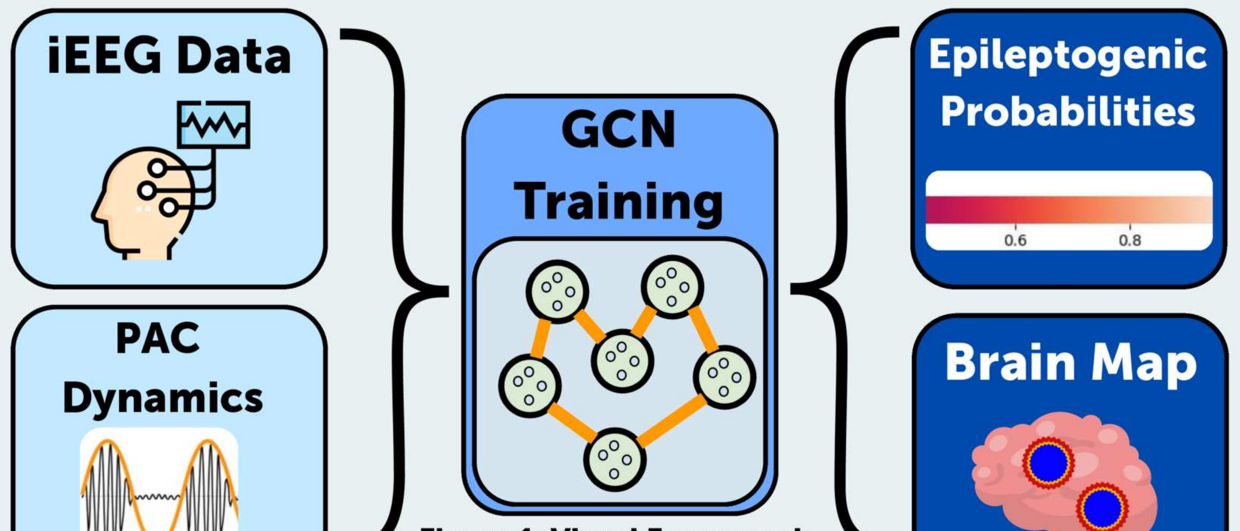


Figure 1: Visual Framework
Flowchart by Researcher using Canva, 2026

INTRODUCTION

Drug-Resistant Epilepsy (DRE) Treatment

Drug-resistant epilepsy affects **~15 million people** (WHO, 2024)
Treatment often uses **resective surgery** on the **epileptogenic zone** (EZ), but locating the EZ is inaccurate, with **30% of patients** having failed surgery (Liu & Li, 2022).

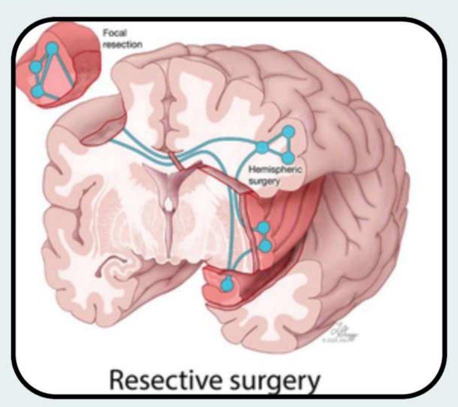


Figure 2: Resective Surgery Procedure (Niazi et al., 2024)

Intracranial EEG (iEEG) Analysis

- Clinicians use **iEEG/fMRI biomarkers** to localize the epileptogenic zone (EZ), motivating more automated and accurate methods (e.g., fMRI connectivity graphs for an **AUC of 0.73**) (Nandakumar et al., 2023).
- iEEG encodes biomarkers like **Phase Amplitude Coupling (PAC)** that are elevated in the EZ (**$p<0.01$**) but **aren't being used** in ML models (Ma et al., 2021).

iEEG Recording → Clinical Analysis → Resective Surgery

RESEARCH OBJECTIVES

Engineering Problem: Resective surgery for DRE is often inaccurate (>30% not seizure-free), and PAC is a promising biomarker that ML models don't use.
Research Question: Can PAC from iEEG, using a Graph Convolutional Network (GCN), accurately localize the EZ?

PAC Creation	***Graph Modeling***	***Brain Maps***
Low & High Frequency Modulation	**Graph** modeling **Network Similarity**	**EZ Probability Heatmaps**

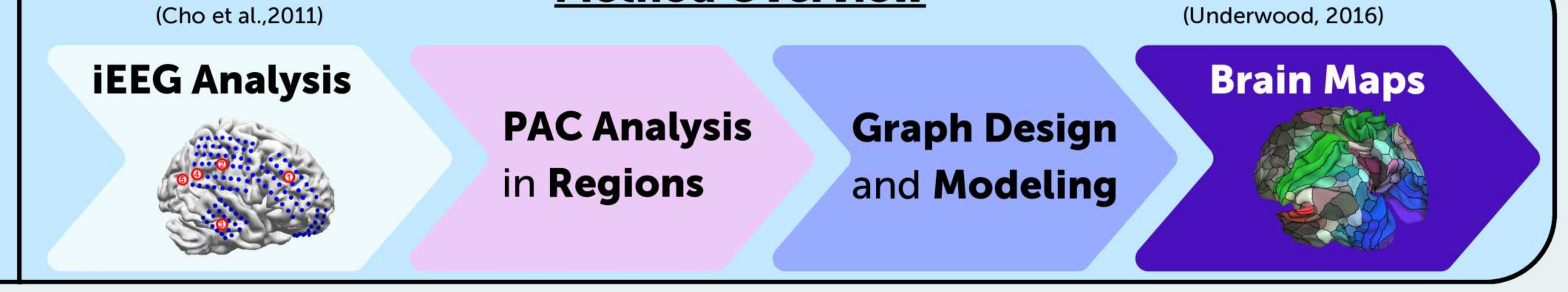
METHODOLOGY

iEEG Data (Zhang et al., 2024)

- **Train Site:** Children's Hospital of Michigan (135)
- **Test Site:** UCLA Mattel Children's Hospital (50)
- **Positive** surgery patients (ages 18-21)

Method Overview

Figure 3: iEEG Placement (Cho et al.,2011)
Figure 4: Sample Brain Map (Underwood, 2016)



1 iEEG Loading & Preprocessing

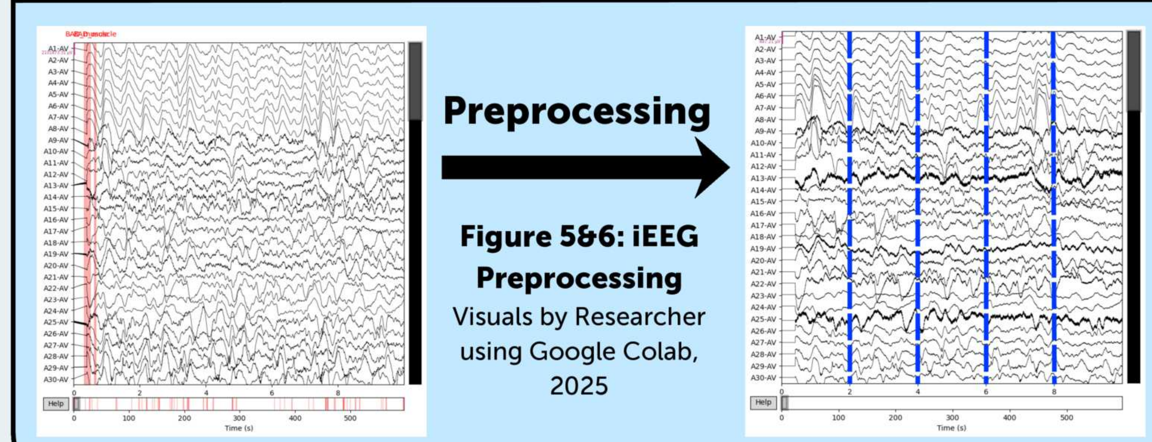


Figure 5&6: iEEG Preprocessing
Visuals by Researcher using Google Colab, 2025

EEG Loaded and **artifacts** removed, split into **windows**

2 PAC Feature Creation

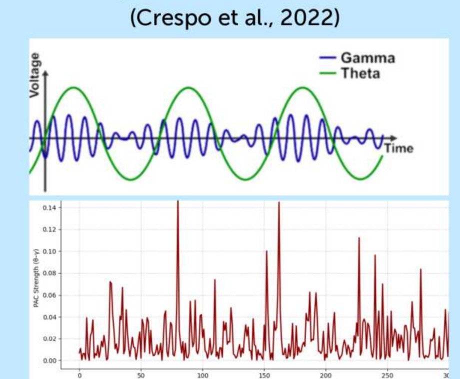


Figure 7: PAC Visual (Crespo et al., 2022)

- Temporal **Theta-Gamma PAC** used for statistical summary
- Electrodes **consolidated**

Consolidation

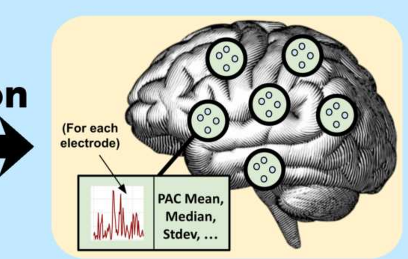
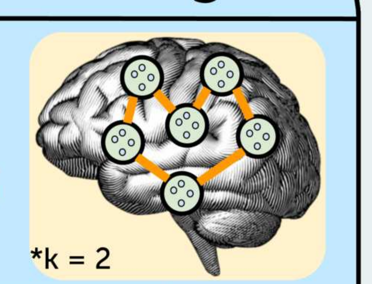


Figure 8: Temporal PAC over Windows
Graph by Researcher using Google Colab, 2026

Figure 9: Regions with PAC
Diagram by Researcher using Google Draw, 2026

3 Graph Forming & GCN Modeling

- A k-nearest neighbor (**kNN**) graph (**k=6**) was constructed
- **Nodes** were **regions**; **edges** encoded similarity in PAC between nodes



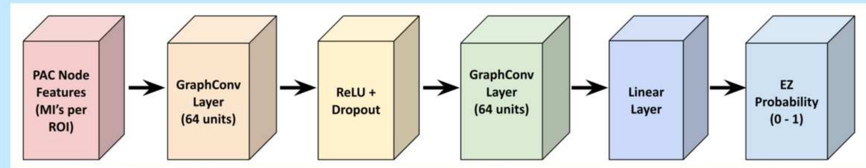


Figure 10&11: Graph Structure and Architecture (Respectively)
Diagrams by Researcher using Google Draw, 2026

4 EZ Probability Heatmap

roi	pred_prob	ez_roi
temporalpole_L	0.992449	1
inferiortemporal_L	0.985382	1
entorhinal_L	0.983254	1

Figure 12: Probability Outputs
Visual by Researcher using Google Colab, 2026

Figure 13: Sample Brain Heatmap
Visual by Researcher using Google Colab, 2026

- The GCN outputs a **probability score (0-1)** of a region's **likelihood of being epileptogenic**
- Probabilities are mapped onto a patient-specific **heatmap** for **EZ visualization**

RESULTS & VALIDATION

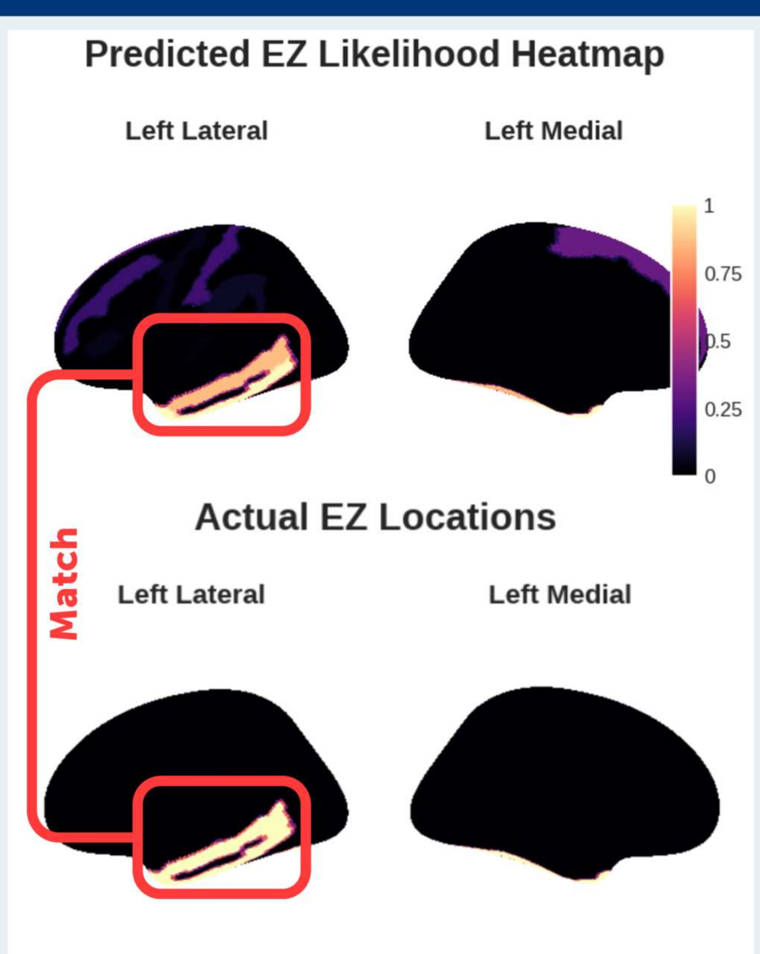


Figure 14: UCLA-Patient1: Probability Heatmap of Predicted EZ Regions vs. Actual EZ Regions
Visual by Researcher using Google Colab, 2026

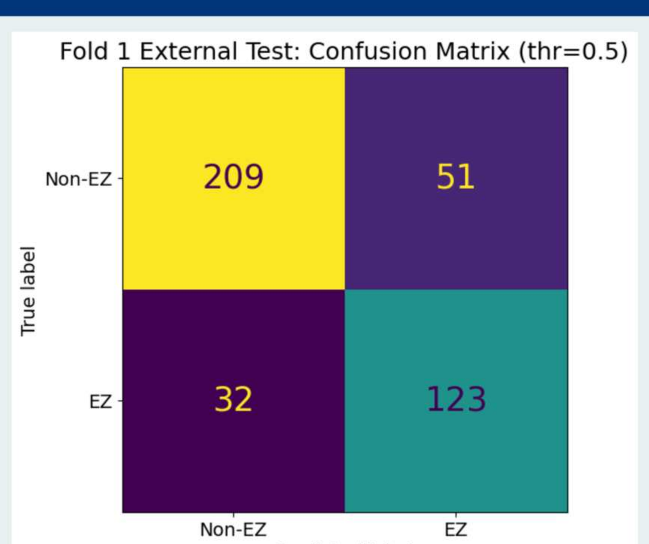


Figure 15: Confusion Matrix of GCN Predictions on Test Set
Visual by Researcher using Google Colab, 2026

- **Model Performance:** High AUC (0.886), high accuracy (0.822), and low false predictions meet design goals
- **Predicted EZ regions** on heatmap **correlate directly** with **actual regions**
- **Hit@1 of 99%** indicates **Top Ranked Region is an EZ 99% of the time**

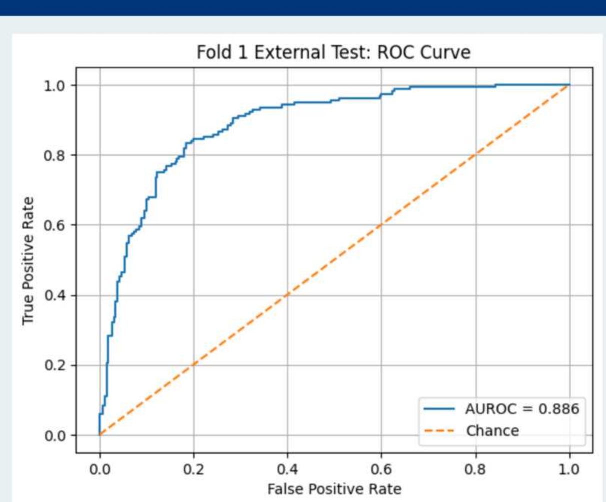


Figure 16: AUROC Curve of GCN Model
Visual by Researcher using Google Colab, 2026

Table 1: Compiled Metrics of GCN Model vs. Control
Visual by Researcher using Google Sheets, 2026

Compiled Metrics of EZ Localization of Models			
Set	**Accuracy**	**AUC**	**F1**
Validation	0.854	0.914	0.741
Test (GCN)	**0.822**	**0.886**	**0.777**
Test (Control)	0.625	0.710	0.594

Hit@1 of 99%

CONCLUSIONS

The **GCN Model** trained on **PAC features** performed **significantly better** than the logistic regression and fMRI model (Nandakumar et al., 2023) in finding the EZ (based on **AUC**).

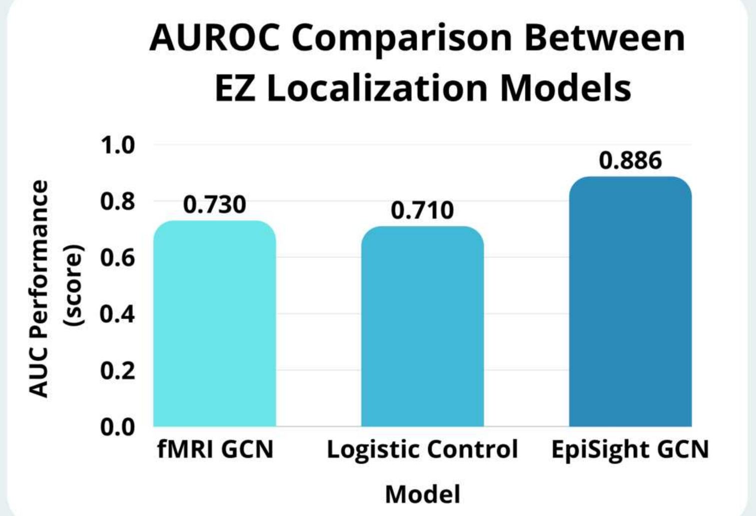


Figure 17: Comparison of EZ Localization Models
Graph by Researcher using Canva, 2026

This model showed that **PAC**, as a sole biomarker, can accurately **localize the epileptogenic zone** through **network level modeling**, due to a **AUC** of ~0.89 and **Hit@1** of 0.99. While this is aligned with previous statistical results (Ma et al., 2021), this study is the **first** to create an **ML model based on PAC**. Combined with a **brain map** of **likely EZ regions**, this could significantly **improve surgery.**

FUTURE WORK

SHAP Feature Isolation

Use **SHAP** to isolate **most impactful PAC features** (mean, median, ...) for future studies

Multi-Biomarker Integration

Combine PAC with **other biomarkers**, including HFOs and fMRI data

Clinical EZ Localization Interface

Construct a **usable clinical interface** for doctors to upload iEEG time series data and output predicted epileptogenic zone probabilities, **guiding surgery**

KEY REFERENCES

Cho, J.-H., Hong, S. B., Jung, Y.-J., Kang, H.-C., Kim, H. D., Suh, M., Jung, K.-Y., & Im, C.-H. (2011). Evaluation of Algorithms for Intracranial EEG (iEEG) Source Imaging of Extended Sources: Feasibility of Using iEEG Source Imaging for Localizing Epileptogenic Zones in Secondary Generalized Epilepsy. *Brain Topography*, *24*(2), 91–104. https://doi.org/10.1007/s10548-011-0173-2

Crespo, L. P., Papaioannou, O., Clark, K., Ogbuagu, N. N., Alliende, L. M., Silverstein, S. M., & Erickson, M. A. (2022). Is Cortical Theta-Gamma Phase-Amplitude Coupling Memory-Specific? *Brain Sciences*, *12*(9), 1131. https://doi.org/10.3390/brainsci12091131

Liu, Y., & Li, C. (2022). Localizing targets for neuromodulation in drug-resistant epilepsy using intracranial EEG and computational model. *Frontiers in Physiology*, *13*, 1015838–1015838. https://doi.org/10.3389/fphys.2022.1015838

Ma, H., Wang, Z., Li, C., Chen, J., & Wang, Y. (2021). Phase–Amplitude Coupling and Epileptogenic Zone Localization of Frontal Epilepsy Based on Intracranial EEG. *Frontiers in Neurology*, *12*. https://doi.org/10.3389/fneur.2021.718683

Nandakumar, N., Hsu, D., Ahmed, R., & Venkataraman, A. (2023). DeepEZ: A Graph Convolutional Network for Automated Epileptogenic Zone Localization From Resting-State fMRI Connectivity. IEEE Transactions on Biomedical Engineering, 70(1), 216–227. https://doi.org/10.1109/tbme.2022.3187942

Niazi, F., Han, A., Stamm, L., Shlobin, N. A., Korman, C., Hoang, T. S., Kielian, A., Du Pont-Thibodeau, G., Ducharme Crevier, L., Major, P., Nguyen, D. K., Bouthillier, A., Ibrahim, G. M., Fallah, A., Hadjinicolaou, A., & Weil, A. G. (2024). Outcome of emergency neurosurgery in patients with refractory and super-refractory status epilepticus: a systematic review and individual participant data meta-analysis. *Frontiers in Neurology*, *15*. https://doi.org/10.3389/fneur.2024.1403266

Underwood, E. (2016). Updated human brain map reveals nearly 100 new regions. Science. https://doi.org/10.1126/science.aag0675

WHO. (2024). *Epilepsy*. World Health Organization. https://www.who.int/news-room/fact-sheets/detail/epilepsy

Zhang, Y., Daida, A., Liu, L., Kuroda, N., Ding, Y., Oana, S., Monsoor, T., Duan, C., Hussain, S. A., Qiao, J. X., Salamon, N., Fallah, A., Sim, M. S., Sankar, R., Staba, R. J., Engel, J., Jr., Asano, E., Roychowdhury, V., & Nariai, H. (2024). *Open iEEG Dataset (Version 1.0.1) [Data set]*. OpenNeuro. https://doi.org/10.18112/openneuro.ds005398.v1.0.1